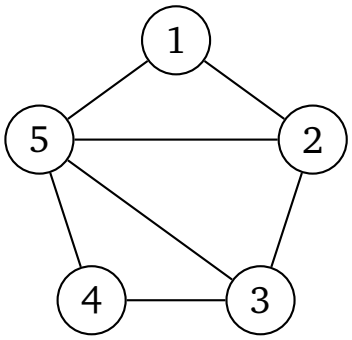
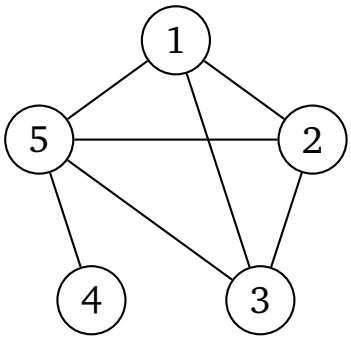
Part 1 — How imperfect can a group be?

Here are two study groups of five. In neither group is everybody friends with everybody. Each group has seven friendships.



the Monday group



the Friday group

Question 1: Without counting anything, which of the two looks more like a real group to you? Say why in one sentence.

Question 2: Work out four numbers for each group and write them in the table.

	the fewest friends any member has	the most members any one person is *not* friends with	friendships, out of the 10 possible	the most steps between two members
name				
Monday				
Friday				

Each column is one of the four definitions from the lecture: k-core, k-plex, ρ-dense and n-clique. Write the right name in the "name" row.

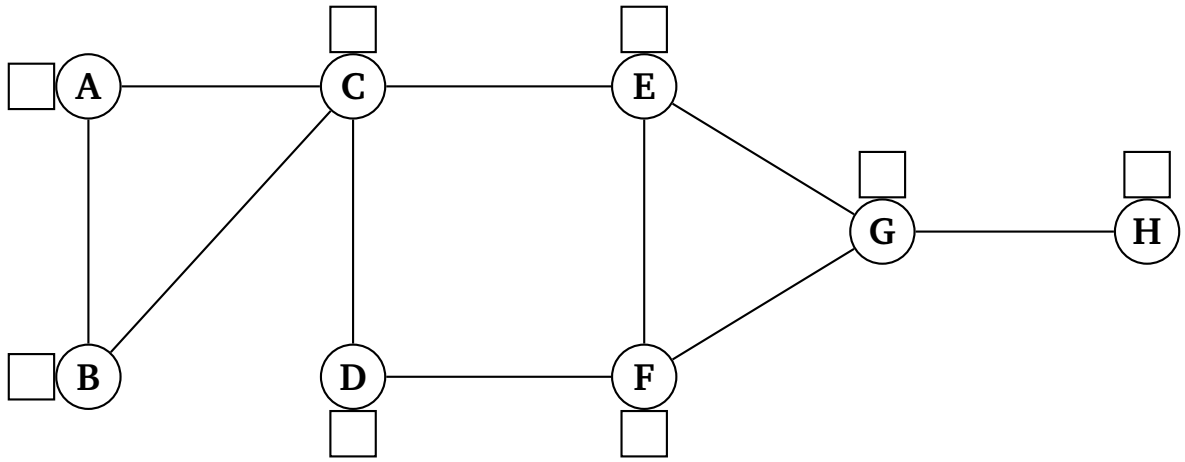
Question 3: Which columns say the Monday group and the Friday group are equally good groups?

Which columns tell them apart?

Look at person 4 in the Friday group. Is person 4 part of that group? Which column would you use to decide? Does your neighbour pick the same one?

Part 2 — Two teams

You teach a lab section of eight students, A to H. The lines are the ten friendships between them. You want to split the section into two teams and keep friends together. The boxes are for Part 3; leave them empty for now.



Question 4: Draw one line on the drawing that splits the eight students into two teams. Cross as few friendships as you can.

How many friendships does your line cross?

Question 5: Compare with your neighbour. Is there a line that crosses even fewer friendships?

Would you really use that line to make two teams? Why?

Question 6: Here are two ways to split the section.

Split X: H alone, and the other seven together.

Split Y: A, B, C, D in one team, and E, F, G, H in the other.

For each split, count the friendships it crosses. Then work out the ratio cut from the lecture:

$$\text{ratio cut} \;=\; \frac{\text{friendships crossed}}{(\text{people in team 1}) \times (\text{people in team 2})}$$

A lower ratio cut is a better split. Which split wins now? What did dividing by the two team sizes change?

Part 3 — More than chance

Play the game from the lecture. Every friendship is a string with a ball on each end, so the ten friendships in Part 2 make twenty balls. A student with three friends holds three balls. Paint each ball in the colour of its owner's team.

Question 7: In each box on the drawing in Part 2, write how many friends that student has.

Question 8: Score four ways to make teams. For each row:

- *observed*: friendships inside a team, divided by 10.
- *chance*: for each colour, take (balls of that colour ÷ 20), square it, and add these up over the colours.
- Q = observed − chance.

teams	friendships inside a team	balls of each colour	observed	chance	Q
everybody in one team					
Split X: {H}, {the rest}					
Split Y: {A,B,C,D}, {E,F,G,H}					
{A,B,C,D}, {E,F,G}, {H}					

Question 9: Which row has the highest Q?

Split X crossed the fewest friendships in Part 2. What does Q say about it?

The ratio cut in Question 6 had to be told "two teams". Did anybody tell Q how many teams to make?

Question 10: Without calculating: put every student in a team of their own, eight teams of one. What is *observed*? Is Q positive or negative? Why?